psnr Function Test Results

Test Case 1: Identical Images

Matrix A:

10	20
30	40

Matrix B:

10	20
30	40

PSNR: Inf

Test Case 2: All Zeros

Matrix A:

0	0	0
0	0	0
0	0	0

Matrix B:

0	0	0
0	0	0
0	0	0

PSNR: Inf

Test Case 3: Constant Matrices

Matrix A:

50 50 50

50 50 50

50 50 50

Matrix B:

100 100 100

100 100 100

100 100 100

PSNR: -33.979

Test Case 4: Negative Values

Matrix A:

-1 -2

3 4

Matrix B:

1 2

-3 -4

PSNR: -14.771

Test Case 5: Floating Values

Matrix A:

0.1000 0.2000

0.3000 0.4000

Matrix B:

0.1000 0.2500

0.3500 0.4500

PSNR: 27.270

Test Case 6: Single Element

Matrix A:

5

Matrix B:

10

PSNR: -13.979

Test Case 7: Random Matrix

Matrix A:

0.6431	0.1986	0.2045
0.6784	0.9587	0.6613
0.8293	0.6625	0.6356

Matrix B:

0.545460	0.045636	0.658498
0.973539	0.058985	0.448681
0.840167	0.735889	0.218786

PSNR: 8.45

Test Case 8: Custom Peak Value (128)

Matrix A:

10	20
30	40

Matrix B:

12	18
33	39

PSNR: 35.612

Test Case 9: High Dynamic Range

Matrix A:

1000 2000

3000 4000

Matrix B:

1100 1900

3100 3900

PSNR: -40

Test Case 10: Binary Image

Matrix A:

0 1

1 0

Matrix B:

1 0

0 1

PSNR: 0

Test Case 11: Mixed Sign Values

Matrix A:

-10 0

10 20

Matrix B:

10 0

-10 -20

PSNR: -27.782

Test Case 12: Large Random Matrix

Matrix A Size:

50 50

Matrix B Size:

50 50

PSNR: -40.238

SUMMARY TABLE

TC	Description	Expected Output	Obtained Output
1	Identical Images	∞	∞
2	All Zeros	∞	∞
3	Constant Matrices	-33.979 dB	-33.979 dB
4	Negative Values	-14.771 dB	-14.771 dB
5	Floating Values	27.27 dB	27.27 dB
6	Single Element	-13.979 dB	-13.979 dB
7	Random Matrix	Finite Value	8.45 dB
8	Custom Peak Value	35.612 dB	35.612 dB
9	High Dynamic Range	-40 dB	-40 dB
10	Binary Image	0 dB	0 dB
11	Mixed Sign Values	-27.782 dB	-27.782 dB
12	Large Random Matrix	Finite Value	-40.333 dB